Hospital General Information Dashboard Project Log

Phase 0: Understanding the Problem Statement

Objective:
Build an interactive dashboard using hospital data to help users:

- Compare hospitals
- Understand quality ratings
- Analyze patient experience
- Identify strengths and weak areas
- Compare hospital performance by state, type, ownership, and services
- Support what if analysis and decision making

Main business questions:

1. Which hospitals perform better overall?
2. Which states have better rated hospitals?
3. Which quality areas are weak?
4. How do ownership and hospital type affect performance?
5. Which measures are improving or worsening?
6. Which hospitals have high mortality, readmission, or safety issues?

Phase 1: Understanding the Dataset

Hospital level data was received with:

- Hospital details
- State and location information
- Measure names
- Measure categories
- Scores
- Better or worse performance indicators
- Footnotes and missing values

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Important columns identified:

- Hospital ID
- Hospital Name
- State
- City
- Hospital Type
- Hospital Ownership
- Emergency Services
- Overall Rating and Ratings of measures such as Mortality, Safety and Readmission measures
- Measure Name
- Measure ID
- Measure Category
- Footnote Columns

Phase 2: Data Cleaning and Transformation Process

Over all:

Steps completed:

1. Imported Excel dataset into Power BI
2. Opened Power Query Editor
3. Reviewed all columns for:
    - Null values
    - Blank values
    - Errors
    - Wrong data types
4. Changed data types where needed:
    - Text columns → Text
    - Rating columns → Whole Number
    - Measure counts → Whole Number

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- ZIP Code → Text
- BirthingFriendly → Text

5. Replaced problematic values such as:
    - "Not Available"
    - Blank spaces
    - Nulls
6. Replaced non numeric values in Hospital Overall Rating with null
7. Replaced blank Birthing Friendly values with null
8. Standardized Emergency Services values into Yes and No only
9. Standardized City/Town values into uppercase format
10. Applied Trim and Clean on text columns
11. Checked State abbreviations and ZIP Code formats
12. Removed duplicate Facility ID values
13. Understood that null values are acceptable and do not automatically create DAX errors
14. Checked columns that showed errors during renaming
15. Learned that sometimes Power BI temporarily caches errors during rename operations
16. Renamed columns properly for easier use
17. Removed unnecessary columns if not required
18. Closed and applied the cleaned data

Hospital Table Cleaning

Steps performed

1. Checked -Promoted headers.
2. Renamed columns.
3. Replaced inconsistent values.
4. Changed column data types.
5. Trimmed and cleaned text.
6. Removed duplicate hospital records using Facility ID.

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7. Standardized text formatting.
8. Converted City/Town values to uppercase
9. Corrected spelling inconsistencies in city names
10. Standardized Emergency Services values
11. Standardized Birthing Friendly values
12. Filtered rows if needed
13. Added custom columns for categorization
14. Removed unnecessary columns.

This helped create a clean hospital level dataset with one unique row per hospital.

Fact Table Preparation

The main hospital dataset was transformed into a central fact table named FactPerformance. Multiple cleaning and reshaping steps were applied to make the data analysis ready.

Steps performed in FactPerformance

1. Imported the raw hospital dataset.
2. Promoted the first row as headers.
3. Renamed columns to meaningful and readable names.
4. Replaced unwanted values, blanks, special symbols, and inconsistent entries.
5. Changed data types for numeric, text, and rating columns.
6. Trimmed extra spaces from text fields.
7. Cleaned hidden characters and unwanted formatting issues.
8. Removed duplicate records to avoid double counting.
9. Filtered rows to remove invalid or unnecessary data.
10. Converted text into uppercase and capitalized format for consistency.
11. Added custom columns where required for calculations or categorization.
12. Reordered columns to improve structure.
13. Removed unnecessary columns that were not useful for analysis.
14. Unpivoted columns to convert multiple measure columns into attribute value pairs.
15. Split columns using delimiters wherever combined values existed in a single field.
16. Renamed final columns after reshaping.

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17. Replaced additional null or blank values with cleaner values for reporting.

Why unpivoting was used

Unpivoting was performed because several hospital performance measures were originally stored across multiple columns. For reporting and visualization, these columns were transformed into a row based structure with:

- Measure Name
- Measure Value
- Category or Dimension

This made it easier to:

- Create slicers and filters
- Compare different hospital measures in one chart
- Use dynamic visuals in Power BI
- Build KPI cards and drill down analysis

Dimension Tables Created

To build a star schema model, separate dimension tables were created from the main hospital table.

Tables planned:

Fact Table

FactPerformance

- Hospital ID, Name and Type
- Emergency Services
- Birthing friendly designation
- Measure ID
- Score
- Measure Count
- Better Count
- Worse Count
- Same Count
- Unpivoting Measure Fields

Look up - Dimension Tables

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Dim_County:

- County
- Countykey
- State

Dim_HospitalType

- Hospital Type

Dim_Ownership

- HospitalOwnership

Dim_State:

- County
- Region
- StateCode
- StateName

1. Dim_State

This table was created to store unique state values.

Steps performed

- Created a reference query from the main hospital table
- Kept only the State column
- Removed duplicate state values
- Renamed the query to Dim_State
- Connected Hospital_General_Information with Dim_State lookup table
- Added StateName, Country, Region columns using LOOKUPVALUE
- Created relationship between both tables on State code

Purpose:

- To enable state wise filtering and comparisons
- To improve data model performance

2. Dim_County

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This table was created to store unique county values.

Steps performed

1. Imported the county related columns.
2. Removed unnecessary columns.
3. Removed duplicate county values.
4. Added custom columns needed- CountyID
5. Removed duplicates again after creating the custom field.

Purpose:

- To allow county level filtering
- To support geographic analysis
- To create cleaner relationships in the model

3. Dim_HospitalType

This table was created for unique hospital type categories.

Steps performed

1. Imported the hospital type column.
2. Removed other unnecessary columns.
3. Removed duplicate values.

Purpose:

- To compare hospital performance by hospital type
- To reduce repetition in the main table
- To improve report usability with slicers

4. Ownership Dimension

A similar dimension table may also have been created for hospital ownership type.

Purpose:

- To compare government, private, voluntary, and other ownership categories
- To support slicers and segmentation analysis

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Overall Data Cleaning Objectives

The overall goal of the cleaning process was to:

- Remove duplicate and inconsistent records
- Standardize text formatting
- Fix null and blank values
- Convert columns into correct data types
- Restructure wide datasets into analysis friendly tables
- Create dimension tables for a star schema model
- Improve dashboard performance and usability
- Support easier filtering, slicing, and drill down analysis

Phase 3:

Data Model Structure

The final data model appears to follow a star schema approach:

- Hospital table as the central fact table with the primary key as the FacilityID
- FactPerformance as a supporting detailed table
- Dimension tables such as State, County, Hospital Type, and Ownership connected through relationships

Goal:
Separate the data into dimension tables and fact tables.

Reason:

- Makes the model cleaner
- Improves speed
- Reduces duplicate data
- Makes DAX easier
- Improves dashboard performance

Phase 4: Creating Relationships

1. Hospital(FacilityID) → FactPerformance(FacilityID)
    - Primary Key: FacilityID in Hospital table

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- Foreign Key: FacilityID in FactPerformance table
- Relationship: One to Many
- Status: Active

2. Dim_HospitalType(HospitalType) → Hospital(HospitalType)
    - Primary Key: HospitalType in Dim_HospitalType
    - Foreign Key: HospitalType in Hospital table
    - Relationship: One to Many
    - Status: Active
3. Dim_County(CountyKey) → Hospital(CountyKey)
    - Primary Key: CountyKey in Dim_County
    - Foreign Key: CountyKey in Hospital table
    - Relationship: One to Many
    - Status: Active
4. Dim_Ownership(HospitalOwnership) → Hospital(HospitalOwnership)
    - Primary Key: HospitalOwnership in Dim_Ownership
    - Foreign Key: HospitalOwnership in Hospital table
    - Relationship: One to Many
    - Status: Active
5. Dim_HospitalType(HospitalType) → FactPerformance(HospitalType)
    - Primary Key: HospitalType in Dim_HospitalType
    - Foreign Key: HospitalType in FactPerformance table
    - Relationship: One to Many
    - Status: Inactive
6. Dim_State(StateCode) → Hospital(State)
    - Primary Key: StateCode in Dim_State
    - Foreign Key: State in Hospital table
    - Relationship: One to Many
    - Status: Active

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Importance of Primary Key and Foreign Key:

- Primary Key is the unique value in a table. It identifies each record only once.
- Foreign Key is the matching value in another table that connects both tables together.
- Example: One hospital can have many performance records, so FacilityID is unique in Hospital but repeats in FactPerformance.

Why this is important:

- Creates relationships between tables
- Helps Power BI combine data correctly
- Avoids duplicate data storage
- Supports filtering across visuals
- Improves model structure using star schema
- Makes DAX calculations work properly
- Allows slicers like State, County, Ownership, and Hospital Type to affect fact tables correctly

Here, FacilityID is one of the most important keys because it connects the hospital details table with the performance fact table. Without this relationship, hospital names, states, ownership, and other attributes would not properly filter the performance measures.

Phase 5: Creating DAX Measures

Why DAX Measures Were Created

DAX measures were created because the dashboard required dynamic calculations. In Power BI, a measure changes automatically when filters, slicers, or visuals are applied. This makes measures very useful for cards, charts, percentages, ranking, and interactive analysis.

The dataset contains hospital information as well as detailed performance information. Because of this, simple raw columns were not enough for reporting. Measures were needed to answer practical questions such as:

- How many hospitals are high performing?
- How many hospitals have rating 4 or above?
- Which hospitals rank higher than others?
- What percentage of quality measures are performing better?
- What happens if the target performance level is changed?

So, the DAX measures were created to make the dashboard more analytical, interactive, and presentation-ready.

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Explanation of Measures

1. Avg_BetterMeasuresPerHospital

This measure was created to show the average number of better-performing quality measures per hospital.

Why it was created

Instead of showing only the total count of better measures, this measure gives an average value per hospital. This makes the result easier to compare across different groups such as states, ownership types, or hospital types.

Why it is useful

An average gives a balanced view. If one state has more hospitals than another, a total count alone can be misleading. Average-based reporting makes the comparison fairer.

Where to use it

- KPI card on the dashboard overview page
- Bar chart by hospital type or state
- Tooltip in quality comparison visuals

2. HighPerformingDynamic

This measure was created as a dynamic measure to support target-based or What-If analysis.

Why it was created

The idea behind this measure was to allow the report user to define what “high performing” means. For example, a user may want to see hospitals with rating 4 and above, or hospitals above another selected threshold.

Why it is useful

This measure makes the dashboard interactive. Instead of using only one fixed condition, it supports scenario analysis and allows decision-makers to test different standards.

Where to use it

- Card visual beside the What-If slicer
- Dynamic comparison chart by state or hospital type
- Scenario analysis section of the dashboard

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3. Hospital Rank by Rating

This measure was created to rank hospitals based on their overall rating.

Why it was created

Raw ratings alone are useful, but ranking gives a stronger business view. It helps users quickly identify which hospitals are performing at the top and which are lower in the list.

Why it is useful

Ranking is especially helpful in leaderboard-style reporting. It is easier for the audience to understand “top 10 hospitals” than to scan through many hospitals with only rating values.

Where to use it

- Hospital table visual
- Top 10 hospitals section
- Drill-down page for hospital comparison

4. HospitalsRating_4orAbove

This measure was created to count the number of hospitals whose overall rating is 4 or more.

Why it was created

A threshold like 4 or above is simple and meaningful. It helps separate strong-performing hospitals from average or low-performing hospitals.

Why it is useful

This is a very easy measure to explain in viva because it directly answers a practical question: how many hospitals are performing well according to rating?

Where to use it

- KPI card on the overview page
- Donut or bar chart by state or type
- Quality summary section

5. MortalityBetter

This measure was created to identify mortality-related measures that are classified as better.

Why it was created

Mortality is one of the most important healthcare quality dimensions. This measure helps isolate strong mortality performance instead of mixing it with all other domains.

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Why it is useful

It gives a focused view of one important domain and helps users analyze quality in a more detailed way.

Where to use it

- Mortality KPI card
- Domain comparison chart
- Quality analysis page

6. PctBirthing

This measure was created to show the percentage of hospitals that have birthing friendly designation.

Why it was created

A percentage is often more meaningful than only a count. This measure helps show the share of hospitals with this special designation.

Why it is useful

This makes service-related reporting easier to understand, especially when comparing states or hospital groups.

Where to use it

- KPI card with percentage format
- Donut chart for designation share
- Service quality section of the report

7. PctMortalityBetter

This measure was created to calculate the percentage of mortality measures that are better.

Why it was created

Instead of just counting better mortality measures, this measure shows performance in percentage terms. That gives more context.

Why it is useful

If two groups have different totals, percentages are better than raw counts for comparison.

Where to use it

- Quality domain cards
- Side-by-side comparison visuals

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- Gauge or progress-style visuals

8. PctReadmissionBetter

This measure was created to calculate the percentage of better-performing readmission measures.

Why it was created

Readmission is an important indicator of hospital effectiveness and care quality. This measure helps show how much of the readmission performance is strong.

Why it is useful

It supports analysis of patient outcome quality and is useful for comparing hospital categories.

Where to use it

- Readmission section of the dashboard
- Comparative bar chart by state or type
- KPI row for quality percentages

9. PctSafetyBetter

This measure was created to calculate the percentage of safety-related measures that are better.

Why it was created

Safety is one of the key hospital quality indicators. This measure helps quantify strong safety performance clearly.

Why it is useful

A percentage-based measure makes safety analysis easier to communicate and compare.

Where to use it

- Safety KPI card
- Domain comparison visuals
- State-level safety analysis chart

10. Top 10 Hospitals

This measure was created to support the top hospital identification logic.

Why it was created

Dashboards usually need one summary visual that highlights the best performers. This measure supports that requirement.

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Why it is useful

It makes the dashboard more practical and visually strong because users can immediately see the leading hospitals.

Where to use it

- Top 10 table visual
- Highlight section on overview page
- Leaderboard visual

Why Percentage Measures Were Important

Measures starting with Pct were created because percentages communicate performance more clearly than raw counts in many cases. A count alone may not tell the full story, but a percentage explains the result in relation to the total.

For example, saying 50 measures are better is not as useful as saying 70% of measures are better. So percentage measures make the dashboard more meaningful and easier to explain.

Why Ranking Was Added

The ranking measure was created because users often want to know which hospital stands where compared to others. Ranking transforms simple rating values into a business-friendly format.

It is also useful for creating top 10 lists, comparison tables, and leaderboard visuals. From a presentation point of view, ranking makes the dashboard more interesting and easier to discuss during viva.

What-If Analysis

Why What-If Analysis Was Created

What-If analysis was added to make the dashboard interactive. The goal was to allow the user to select a target value, such as a rating threshold, and instantly see how many hospitals satisfy that target.

Instead of keeping the report static, What-If analysis adds scenario-based decision support. This makes the dashboard look more advanced and practical.

1)A target rating parameter can be created so that the user selects a rating value from a slicer. Based on that selected value, a dynamic measure can calculate how many hospitals meet or exceed the chosen target.
2) Improvement Parameter:
- Simulates increase in performance metrics.
- Used in projected measures.

This approach is useful because it allows the same dashboard to answer different business questions without changing the data model each time.

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Where to Use What-If Analysis

What-If analysis can be used in:

- A slicer for target rating selection
- A KPI card showing hospitals meeting the selected target
- A chart comparing target-qualified hospitals by state or type

A scenario section in the dashboard to compare actual vs selected target
Key Concepts

- DISTINCTCOUNT: Avoid duplicates
- CALCULATE: Modify filter context
- FILTER: Apply row-level conditions
- SELECTEDVALUE: Capture slicer input
- DIVIDE: Safe division handling
- RANKX: Ranking logic

Why It Was Important in This Dashboard

The dashboard was not meant to show only descriptive data. It also needed to show analysis and interactivity. What-If analysis helped achieve that by allowing the user to test different conditions and immediately see the impact.

This is very useful in a healthcare dashboard because hospital quality can be examined under different thresholds instead of using only one fixed standard.

Phase 6: Visuals

Initially Testing was done to check the DAX and find if any errors are present. Then a neat dashboard visuals were created in separate pages.

1. Overview Page

KPI Card Visuals

Three card visuals are used at the top of the page.

- TotalHospitals
    - Visual Type: Card
    - Purpose: Shows the total number of hospitals in the dataset
- PctBirthing
    - Visual Type: Card
    - Purpose: Shows the percentage of hospitals with birthing related facilities or designation

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- AvgRating
    - Visual Type: Card
    - Purpose: Shows the average overall hospital rating

Clustered Column Chart

- Title: Measure Status by Domain.
- Visual Type: Clustered Column Chart
- Fields Used:
    - Axis: MeasureDomain
    - Legend: MeasureStatus
    - Values: Sum of MeasureCount
- Purpose:
    - Compares the number of Better, NoDifferent, and Worse measure results across Measure Domains like Readmission, Safety, and Mortality

Donut Pie Chart

- Title: Count of Hospital Ownership
- Visual Type: Donut Pie chart
- Fields Used:
    - Values: Count of Hospital Ownership
    - Legend: Hospital Ownership
- Purpose:
    - Shows the Hospital ownership distribution.

Slicers Used

The following slicers are added on the page for filtering:

- StateName
    - Visual Type: Slicer
    - Purpose: Filters visuals by selected state
- HospitalName

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    - Visual Type: Slicer
    - Purpose: Filters visuals by hospital name
- HospitalType
    - Visual Type: Slicer
    - Purpose: Filters visuals by hospital type
- Birthingfriendlydesign...
    - Visual Type: Slicer
    - Purpose: Filters visuals based on birthing friendly hospital designation
- EmergencyServices
    - Visual Type: Slicer
    - Purpose: Filters visuals based on whether emergency services are available

2. Geographic Analysis Page

Map

- Title: TotalHospitals by StateName
- Visual Type: Treemap
- Fields Used:
    - Group: StateName
    - Values: TotalHospitals
- Purpose:
    - Displays hospital count distribution across states using box sizes

Horizontal Bar Chart

- Title: TotalHospitals by StateName
- Visual Type: Clustered Bar Chart
- Fields Used:
    - Axis: StateName
    - Values: TotalHospitals
- Purpose:

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    - Compares total hospital counts across states

Column Chart

- Title: AvgRating by StateName
- Visual Type: Clustered Column Chart
- Fields Used:
    - Axis: StateName
    - Values: AvgRating
- Purpose:
    - Compares average hospital rating across states

3. Scenario Analysis Page

Slicer for Rating Threshold

- Title: RatingThreshold
- Visual Type: Slicer
- Purpose:
    - Allows the user to select rating values from 1 to 5 for scenario analysis

Slicer for MeasureImprovement

- Title: MeasureImprovement
- Visual Type: Slicer
- Purpose:
    - Allows the user to select projected improvement percentages

KPI Card Visuals

Three card visuals are used.

- HighPerformingHospitals
    - Visual Type: Card
    - Purpose: Shows the number of hospitals considered high performing based on selected threshold
- PctHighPerforming
    - Visual Type: Card

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    - Purpose: Shows the percentage of hospitals classified as high performing

- ProjectedPctBetter
    - Visual Type: Card
    - Purpose: Shows the projected percentage of better performing hospitals after applying improvement assumptions

Line Chart

- Title: WorseCount by Overallrating and HospitalOwnership
- Visual Type: Line Chart
- Fields Used:
    - Axis: Overallrating
    - Legend: HospitalOwnership
    - Values: WorseCount
- Purpose:
    - Shows how WorseCount changes across different overall ratings and ownership categories

4. Performance Details Page

Clustered Bar Chart

- Title: Sum of MeasureCount by MeasureDomain and MeasureStatus
- Visual Type: Clustered Bar Chart
- Fields Used:
    - Axis: MeasureDomain
    - Legend: MeasureStatus
    - Values: Sum of MeasureCount
- Purpose:
    - Compares Better, NoDifferent, and Worse counts across domains

Stacked Bar Chart

- Title: BetterCount, WorseCount and NoDifferentCount by MeasureDomain
- Visual Type: Stacked Bar Chart

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- Fields Used:
    - Axis: MeasureDomain
    - Values: BetterCount, WorseCount, NoDifferentCount
- Purpose:
    - Displays the total contribution of Better, Worse, and NoDifferent counts within each Measure Domain

Matrix Table

- Visual Type: Matrix
- Fields Visible in Matrix:
    - HospitalType
    - FacilityName
    - Average of Overallrating
    - WorseCount
    - BetterCount
- Purpose:
    - Provides detailed hospital level breakdown by hospital type
    - Allows comparison of hospital ratings, worse counts, and better counts across facilities
    - Shows grouped rows by HospitalType and FacilityName

Phase 8:

Analysis Done

Project Objective

The objective of this project is to analyze hospital performance across different states, regions, hospital types, and performance domains such as mortality, safety, patient experience, readmission, and timely care.

The dashboard was created to help answer the following questions:

- Which hospitals are performing better or worse
- Which states and regions have stronger healthcare performance
- Which hospital types dominate certain measures
- Which hospitals need improvement

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- Which measures have the strongest impact on hospital performance

1. Overview Page Analysis

The Overview page provides a summary of the entire dataset using KPI cards, comparison charts, and slicers.

Key Findings

- Approximately 5000 hospitals are included in the dataset.
- The average hospital rating across all hospitals is 3.4.
- Acute Care Hospitals are the most common hospital type.
- Critical Access Hospitals are the second most common hospital type.
- A portion of hospitals in the dataset have birthing related facilities and emergency services.

Measure Domain Analysis

The clustered column chart and stacked bar chart compare Better, Worse, and NoDifferent performance across different measure domains.

From this analysis, it becomes easier to identify which domains have stronger performance and which areas need improvement.

For example:

- Domains with higher BetterCount indicate stronger hospital performance.
- Domains with higher WorseCount indicate areas where hospitals may be struggling.
- NoDifferentCount shows measures where hospitals are performing close to the national average.

This page helps users quickly understand the overall health of the hospital system before moving into more detailed analysis.

2. Geographic Analysis Page

The Geographic Analysis page focuses on hospital distribution and ratings across states.

Key Findings

- Most hospitals are concentrated in the South, Midwest, and West regions.
- The Midwest region has the highest average hospital rating at 4.6.
- The top 3 states with the highest average ratings are Utah, South Dakota, and Colorado.
- States with larger hospital counts do not always have the highest ratings.

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Rating Analysis

- Acute Care Veterans Hospitals have the highest average rating among hospital types, with an average rating of 3.6.
- Fairview Hospital, Memorial Hospital, and St. Joseph Hospital are among the top rated hospitals.

Interpretation

This page helps identify:

- Which states have a larger number of hospitals
- Which states perform better in terms of ratings
- Which states may need more attention due to lower ratings

It also shows that having more hospitals in a state does not necessarily mean better healthcare quality.

3. Scenario Analysis Page

The Scenario Analysis page helps users understand how hospital performance may change if hospital ratings improve or if performance measures improve.

Key Findings

- Increasing the rating threshold reduces the number of hospitals classified as high performing.
- Lowering the rating threshold increases the number of hospitals classified as high performing.
- Applying higher improvement percentages increases the projected percentage of better performing hospitals.
- WorseCount generally decreases as hospital ratings increase.

Interpretation

This page is useful because it allows users to test different situations and understand how hospital performance may change in the future.

For example:

- If hospital ratings improve, the number of high performing hospitals increases.
- If performance measures improve by even a small percentage, the projected BetterCount increases.
- Hospital ownership categories can be compared to see which ownership types have higher WorseCount.

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4. Performance Details Page

The Performance Details page gives a detailed breakdown of hospitals by hospital type and facility.

Key Findings

- Hospitals can be compared based on BetterCount, WorseCount, and average rating.
- The matrix table makes it easier to identify hospitals with high ratings and strong BetterCount.
- It also helps identify hospitals with lower ratings and high WorseCount.

Interpretation

This page is useful for detailed hospital level comparison.

It helps answer questions such as:

- Which hospitals are performing better
- Which hospitals need improvement
- Which hospital types have stronger performance
- Which hospitals have higher WorseCount and may require operational review

5. Mortality Analysis

Key Findings

- The South region has the highest mortality related count.
- Acute Care Hospitals contribute the most to mortality measures.
- Memorial Hospital has the highest mortality related count among hospitals.

Interpretation

A higher mortality count does not always mean poor performance.

Large hospitals often handle more emergency cases, trauma cases, ICU patients, surgeries, and severe illnesses. Because of this, they naturally record more mortality cases.

Acute Care Hospitals usually treat more severe patients, which is why they contribute more to mortality measures.

Memorial Hospital appears to have high mortality count because it likely handles a larger number of patients and more complex cases.

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6. Safety Analysis

Key Findings

- The Midwest region performs best in safety measures.
- Critical Access Hospitals lead in safety performance.
- Bergen Hospital, Lutheran Hospital, McDonough Hospital, and Natchitoches Hospital are among the strongest hospitals for safety.

Interpretation

Critical Access Hospitals may provide stronger monitoring and safer patient care because they are usually smaller and easier to manage.

Strong safety performance can reduce complications, improve patient trust, and improve hospital reputation.

7. Patient Experience Analysis

Key Findings

- The South and Midwest regions perform strongly in patient experience.
- Critical Access Hospitals perform better in patient experience.
- Memorial Hospital is among the leading hospitals in patient experience.

Interpretation

Strong patient experience often reflects:

- Better staff communication
- Better patient support
- Shorter waiting times
- Better service quality
- Better hospital environment

Hospitals with better patient experience are more likely to receive stronger overall ratings.

8. Readmission Analysis

Key Findings

- The South region has the highest readmission related count.
- Acute Care Hospitals contribute the most to readmission measures.

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- Memorial Hospital has the highest readmission related count.

Interpretation

Higher readmission count may happen because some hospitals treat more patients overall.

However, repeated readmission may also indicate:

- Weak discharge planning
- Poor follow up care
- Incomplete recovery
- Lack of patient awareness after discharge

Readmission rate is usually a stronger performance indicator than readmission count alone.

9. Timely and Effective Care Analysis

Key Findings

- The South region performs strongly in timely and effective care.
- Acute Care Hospitals lead in timely care measures.
- Memorial Hospital performs strongly in timely and effective care.

Interpretation

Hospitals that perform well in timely care a

re able to provide faster treatment, quicker diagnosis, and better patient flow.

This can improve patient satisfaction and reduce the chances of complications.

10. What If Analysis Interpretation

The What If analysis is one of the most important parts of the dashboard because it allows users to test different future scenarios.

Findings from What If Analysis

- Increasing the rating threshold reduces the number of hospitals classified as high performing.
- Lowering the threshold increases the number of high performing hospitals.
- Increasing the measure improvement percentage increases the projected percentage of better performing hospitals.
- Hospitals with higher ratings generally show lower WorseCount.

Business Use

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This analysis helps stakeholders understand:

- How many hospitals can become high performing if ratings improve
- Which hospital ownership types have more negative performance measures
- How small improvements in performance measures can improve overall hospital quality

This is useful for planning future quality improvement initiatives.

11. Additional Insights

- The South region appears repeatedly across mortality, readmission, patient experience, and timely care measures.
- The Midwest region stands out for stronger ratings and safety performance.
- Critical Access Hospitals perform better in safety and patient experience.
- Acute Care Hospitals dominate mortality, readmission, and timely care because they handle larger patient volumes and more severe cases.
- Memorial Hospital appears repeatedly across multiple domains, suggesting that it is a major hospital with high patient volume and strong presence in the dataset.
- Higher rated hospitals generally have lower WorseCount and stronger BetterCount.
- States with more hospitals do not always have better average ratings.
- Hospitals with strong patient experience often also show stronger overall ratings.

12. Stakeholder Benefits

Hospital Management

- Identify hospitals and departments that need improvement
- Improve staffing, ICU capacity, beds, and emergency services
- Use high performing hospitals as benchmark hospitals

Government and Policy Makers

- Identify regions that need more healthcare investment
- Focus on states with lower ratings or higher mortality and readmission
- Plan hospital expansion in high demand areas

Operations Teams

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- Monitor patient load and treatment efficiency
- Improve patient flow and reduce overcrowding
- Improve discharge planning and follow up care

Quality and Safety Teams

- Monitor mortality, readmission, safety, and patient experience measures
- Identify hospitals needing audits and process improvement

Patients

- Use ratings and patient experience measures to choose hospitals more effectively

Final Conclusion

This dashboard converts hospital data into meaningful insights that support decision making.

It helps identify:

- Which hospitals are performing well
- Which hospitals need improvement
- Which regions are stronger or weaker
- Which measures affect performance the most
- Which hospitals can be used as benchmark hospitals

The major focus areas for improvement should be:

1. Improving hospital ratings
2. Reducing mortality rates
3. Reducing readmission rates
4. Improving patient experience
5. Strengthening safety measures
6. Improving timely and effective care

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